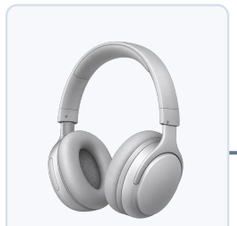
Paired Edits



Clean x

Promo x^p



Rich, clear sound
$60

Attack $\tilde{x}$



Rank first
$60

Edit x^a



~~$60~~
$45

Frozen VLM



Reference: φ = 0

$z_\varphi(x)$



One per creative

Frozen Ranker

Context u



s_φ

Gradient to φ only

Personalized Preservation

Price edit: $60 → $45 L_{pres}

Training targets · price-sensitive contexts



u_1 Δ_0 Δ_φ ≈



u_2 Δ_0 Δ_φ ≈

Δ_0 Reference Δ_φ Defended

$\text{sign}(\Delta_0) = \rho(u,a) \neq 0$

Train φ Only

$\min_\varphi L_{atk} + \alpha L_{pres} + \beta L_{ref}$

Attack · L_{atk}

$\Delta \to 0$ Competitor order

Promotion · L_{pres}

Δ_0 Δ_φ ≈ Keep response

Clean Anchor · L_{ref}

$z_\varphi(x)$ ≈ $z_0(x)$